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Tracking multiple common grounds

Mismatches in activation

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Zusammenfassung

Eine fiktionale Erzählung enthält eine Vielzahl von Common Grounds (CGs). Auf der einen Seite wird ein interner Common Ground durch die Gespräche zwischen den Protagonisten der Erzählung etabliert, und auf der anderen Seite wird ein externer Common Ground für die narrative Information des Erzählers an die Lesenden etabliert. Die Protagonisten der Erzählung haben jedoch keinen Zugang zu diesem externen Common Ground. Der Aktivierungsgrad eines Textreferenten, der die Wahl eines referentiellen Ausdrucks beeinflusst, kann zwischen den CGs variieren. Dieser Beitrag berichtet über ein Experiment, in dem wir Kontexte präsentierten, die zu Mismatches in der Referentenaktivierung zwischen den beiden CGs führten. Die Teilnehmenden sollten dann entweder ein Pronomen oder einen Eigennamen als anaphorischen Ausdruck für ein vorher eingeführten Referenten nutzen, was den jeweiligen Aktivierungsgrad signalisiert. Die Ergebnisse zeigen, dass die Teilnehmenden im Allgemeinen sensibel auf die Unterschiede im Aktivierungsstatus zwischen den CGs reagieren, jedoch häufig die CG der Protagonisten der Erzählung zugunsten der CG des Erzählers/Lesers außer Acht lassen. Dies spricht gegen eine klare Trennung zwischen dem internen und externen CG.

Schlagwörter: Anapher, Pronomen, Aktivierung, Zugänglichkeit, Narrativ, direkte Rede, Redewiedergabe, Perspektive

Abstract

A fictional narrative contains a multitude of common grounds (CGs). On the one hand, the characters talk to each other and develop a CG between them. On the other hand, the narrator is telling the story to the reader and contributes to the narrator/reader CG that the characters are not privy to. The activation level of a referent, which influences the choice of a referring expression, can differ between the CGs. This paper reports an experiment in which we presented participants with contexts that created mismatches in referent activation between CGs and asked them to choose a referring expression, a name or a pronoun, to see which CG determines their referential choice. The results show that readers are generally sensitive to the differences in activation status between the CGs but often disregard the characters' CG in favour of the narrator/reader CG. This speaks against a clean separation between the CG of the reported speech context and that associated with the context of the narration.

Keywords: anaphora, pronouns, activation, accessibility, common ground, narrative, direct speech, quotation, perspective, referential choice

1. Introduction and background

The choice of referring expression depends on the activation of its referent in the current context. Activation (or accessibility), in turn, depends on a number of factors, which Ariel (1990) divides into four major categories:

(1) *Distance*: Closer antecedents make referents more accessible than distant antecedents.
Competition: A referent is more accessible if it has no (or fewer) competitors.
Saliency: A more salient (e.g. topical) referent is more accessible.
Unity: Antecedents within the same unit (frame, world, point of view, discourse segment) make referents more accessible than if they are outside that unit.

Third person pronouns are mostly used to refer to highly accessible/activated antecedents, whereas proper names are used for less activated ones. For instance, in (2) the pronoun *she* is acceptable because its referent Amy is highly activated based on all the four criteria. It has been mentioned in the immediately preceding sentence (short distance to antecedent); there are no other suitable referents in the context that the pronoun could refer to (no competitors); its antecedent is the grammatical subject of the sentence, which is a signal of topicality (saliency); and there is nothing that could be interpreted as a transition from one frame/situation/world to another between the antecedent and the pronoun. The repeated use of a proper name is dispreferred in this context (see also Givón, 1983; Gordon et al., 1993):

(2) "Amy wants to sell the boat," Bill said.
"She/#Amy is asking a fair price, no doubt," I said.

In contrast, if the narrator does not immediately refer to Amy and first talks about the boat, as in (3), the activation of Amy decreases with the growing distance to her first mention. The boat could be seen as a new local discourse topic here (although Amy probably remains the global topic, since it is Amy's boat), so one could argue that a new discourse segment is initiated between that antecedent and the referring expression, breaking discourse segment unity. Therefore, it becomes more acceptable to reintroduce the referent with a proper name, even though pronominal reference with *she* remains possible, since Amy has no female competitors.

(3) "Amy wants to sell the boat," Bill said.
"Finally," I said, picking up the dropped napkin. "The old rust bucket nearly sank twice this year, and the repairs cost half a fortune. I've promised never to set foot in it again. She/Amy is asking a fair price, no doubt."

While it is uncontroversial that the activation of a referent is only defined in a specific context, “context” can mean more than one thing. It obviously has to include previous text. For first and second person reference and other deictic expressions the situational context is also crucial. What is often overlooked is that the situational context, in particular who is speaking to whom, also imposes constraints on third person reference. The ultimate goal of choosing the right referring expression is for the speaker to be understood and have the addressee correctly resolve the reference. Therefore it only makes sense to talk about activation relative to the common ground (CG) of *those specific* communication participants, i.e. the co-text, the communicative situation and any other aspects of the context to the extent to which they are represented in *their* CG. For instance, since the referring expression *she/Amy* occurs in the direct speech of the homodiegetic first person narrator, which is part of their conversation with Bill, the choice to pronominalise or not should depend on the activation of the referent in the CG between Bill and the narrator as a character in the story.

However, a fictional narrative typically contains a multitude of CGs. Each pair of interacting characters will have their shared *internal* CG. On top of that, the narrator is telling the story to the reader and they develop their *external* CG. Although the notion of CG in one-sided written communication raises a number of theoretical problems (see e.g. Semeijn, 2021), the need to produce an understandable message where all referring expressions will be interpreted correctly exists both in face-to-face dialogue and in written monologue and a referent’s activation affects the choice between pronouns and proper names in both modes of communication.

It is useful to note that the relationship between the external and the internal CGs is asymmetric. The reader learns from the narrator about the communication between the characters and in this way develops a partial representation of those characters’ CGs. But the characters (normally) do not know that the narrator is telling a story to a reader and have no representation of the external CG. In other words, the external CG contains a representation of the internal CG, but not the other way around. However, that does not mean that the activation status of referents in the internal CG unilaterally determines their status in the external CG. While mentions (or non-mentions) of a referent in the direct speech of characters affect its status in both the internal and the external CG, mentions outside direct speech only affect the external CG. As we will see presently, that can lead to mismatches in the status of a referent across CGs.

The main question addressed in this paper is what happens if the activation level of a referent in the internal and the external CG is not the same. In (4), for instance, the first mention of Amy activates the referent both in the internal CG between Bill and the first person narrator and in the external CG of the narrator and the reader. However, the whole talk about the boat takes place in the thoughts of the narrator, so it is part of the narrator's message to the reader, but not to Bill. Therefore, by the time Amy is mentioned again, her activation in the narrator/reader CG should be relatively low, presumably about as low as in (3). However, since Bill does not hear the intervening sequence about the boat, at best he might notice a somewhat longer pause (caused by the narrator's thoughts) before the narrator reacts to his utterance, but for Bill, Amy is still the subject of the immediately preceding sentence and no topic change has happened in his representation of the context. So Amy's activation in the narrator/Bill CG should still be relatively high, presumably about as high as in (2). The question is: Will the narrator use the pronoun to cater to Bill's CG management needs, or the proper name to cater to the reader?

(4) "Amy wants to sell the boat," Bill said.
Finally, I thought, picking up the dropped napkin. The old rust bucket had nearly sunk twice this year, and the repairs had cost half a fortune. I had promised never to set foot in it again.
"She/Amy is asking a fair price, no doubt," I said.

The mismatch in referent activation between the external and the internal CG that we see in (4) can also go in the other direction. Example (5) is like (3), except Amy gets mentioned again and reactivated just before the last sentence containing the critical referring expression. However, the reactivation takes place in a sentence that presents another thought of the narrator and Bill does not hear that. So for Bill, the use of a proper name in this context would be more appropriate, while the reader might prefer a pronoun. The question is which one is actually preferred.

(5) "Amy wants to sell the boat," Bill said.
"Finally," I said, picking up the dropped napkin. "The old rust bucket nearly sank twice this year, and the repairs cost half a fortune. I've promised never to set foot in it again."
Amy was cross and hadn't spoken to me since.
"She/Amy is asking a fair price, no doubt," I said.

Answers to these questions bear on a more general theoretical issue: What is context, how is it structured, and what place does it assign to the CG? Direct speech in fictional narrative, which the examples above try to imitate, is a form of quotation. The possibility of anaphoric links that cross the quotation boundary poses a number of problems to theoretical semantics (Partee, 1973; Gasparri, 2019; Maier, 2020; Anderson, 2025). The problem illustrated above adds to that list. If, in the Kaplanian spirit, context is determined by who speaks to whom, then the reference to Amy in the last sentence of (5) and the reference in the immediately preceding sentence happen in different contexts. Therefore, the latter should not count as the antecedent and should not affect the activation of Amy in the last sentence. If readers separately keep track of the different CGs at different narrative layers and form their preferences for referring expressions within the boundaries of the respective CG, the reactivation of a referent in the thoughts of the narrator should have no effect on the choice of referring expression in the speech of a character, even when that character is the narrator, as in the examples above.

Doubts about such a perfect separation between CGs come from a large body of work in psycholinguistics (see e.g. Heller and Brown-Schmidt, 2023, for a recent summary) showing that speakers sometimes disregard mutual knowledge in favour of their own private information when referring to objects. If readers have a similar egocentric approach to handling multiple CGs in a narrative, one might expect that their preferences would be more strongly affected by referent activation in the CG they are directly part of, i.e. the external, narrator-reader CG. Moreover, it is also possible that context is a more abstract structure that integrates the multiple CGs pertaining to different speech situations in a narrative. Those CGs, in turn, may or may not communicate with each other. Like frames, worlds, or points of view, CGs could be just another parameter that plays a role for the unity factor in determining the activation of a referent, cf. (1). Then antecedents in other CGs would have a lesser effect on activation than antecedents within the same CGs, but the effect is no longer expected to be zero.

We devised an experiment to examine referential choice in the direct speech of characters depending on referent's matching or mismatching activation in the internal and the external CG, using a 2×2 design with intervening speech/thought and +/−reactivation as independent variables, and created texts following the schema in table 1. The difference between the intervening speech and thought conditions was whether the sequence that created the distance between the first and the next mention of the target referent (Amy) was presented as direct speech or (free) indirect thought of the narrator, as in examples (3) vs. (4) above. A proper name referring to the target referent (Amy) was always the subject both in the sentence where it was first introduced and in the sentence preceding the target sentence in the +reactivation condition. In the −reactivation condition that sentence did not mention the target referent; instead, *I* was the subject.

Our expectations for the preferred referring expressions are summarised in table 2. In the intervening thought +reactivation condition there is no mismatch, the target referent is highly activated in both CGs, so the pronoun should be the preferred choice in the target sentence. In the intervening speech −reactivation condition, where the referent is relatively deactivated in both CGs, the name should be, if not preferred, then relatively less dispreferred.

The mismatch conditions in the two middle lines of the table should reveal how readers manage conflicting CGs. If readers are able to keep track of the characters' CG as perfectly separate from the narrator-reader CG and always choose a referring expression intended for Bill's ears from Bill's standpoint, then we expect to find a main effect of the intervening speech/thought variable, and no effect of reactivation, because reactivation takes place in a sentence that does not affect Bill's representation of the context. On the other hand, if readers only pay attention to how activated referents are in their own representation, i.e. the activation pattern in the narrator-reader CG dominates over the characters' CGs, then we expect to see an effect of the reactivation variable, and no effect of the speech/thought variable, because no matter if the intervening sequence is presented as speech or thought, the distance to the antecedent is the same for the reader. Finally, if we find a main effect of both variables, this would be an indication that the conflicting activation patterns interfere in the reader's representation of the context and both influence the choice of referring expression to some extent.

2. Method

2.1 Stimuli

Experimental Items

Four versions of 24 texts were created following the pattern in table 1. The target referent and the interlocutor, Amy and Bill, always had distinct genders, so the pronoun in the target sentence was never ambiguous as it always had only one antecedent of a matching gender in the entire text. The antecedent of the target referent (Amy) was always the sentence-initial subject both in the sentence where it was first introduced and in the reactivation sentence in the +reactivation condition, to exclude grammatical role and position as possible confounds (Brennan et al., 1987; Gordon et al., 1993). The target referring expression, name or pronoun, to be selected by the participant, also always occupied the subject position in the target sentence.

intervening speech, −reactivation	**intervening speech, +reactivation**
Bill pushed the plate of scones towards me. “Amy wants to sell the boat,” he said. “Finally,” I said, picking up the dropped napkin. “The old rust bucket nearly sank twice this year, and the repairs cost half a fortune. I’ve promised never to set foot in it again.” I poured milk into the cups and sat next to Bill. “___ is asking a fair price, no doubt,” I said and rolled a warm scone between my cold hands.	Bill pushed the plate of scones towards me. “Amy wants to sell the boat,” he said. “Finally,” I said, picking up the dropped napkin. “The old rust bucket nearly sank twice this year, and the repairs cost half a fortune. I’ve promised never to set foot in it again.” Amy was cross and hadn’t spoken to me since. “___ is asking a fair price, no doubt,” I said and rolled a warm scone between my cold hands.
intervening thought, −reactivation	**intervening thought, +reactivation**
Bill pushed the plate of scones towards me. “Amy wants to sell the boat,” he said. Finally, I thought, picking up the dropped napkin. The old rust bucket had nearly sunk twice this year, and the repairs had cost half a fortune. I had promised never to set foot in it again. I poured milk into the cups and sat next to Bill. “___ is asking a fair price, no doubt,” I said and rolled a warm scone between my cold hands.	Bill pushed the plate of scones towards me. “Amy wants to sell the boat,” he said. Finally, I thought, picking up the dropped napkin. The old rust bucket had nearly sunk twice this year, and the repairs had cost half a fortune. I had promised never to set foot in it again. Amy was cross and hadn’t spoken to me since. “___ is asking a fair price, no doubt,” I said and rolled a warm scone between my cold hands.

Tab. 1: Example of an experimental item in all four conditions. (Mentions of the target referent and speech/thought tags are highlighted here for presentation purposes. In the experiment only the gap in the target sentence was highlighted.)

reactivation	**intervening**	**Separate CGs**	**Narr/reader CG dominance**	**CG interference**
–	speech	name	name	name
–	thought	pronoun	name	?
+	speech	name	pronoun	?
+	thought	pronoun	pronoun	pronoun

Tab. 2: Predicted relative preferences for referring expressions. (Which expression is relatively more likely to be chosen compared to the same expression in other conditions.)

Furthermore, the sentence did not contain any other third person pronouns. Since the combination of a proper name or definite description subject with a pronominal object is dispreferred for independent reasons (Brennan et al., 1987), we wanted to exclude it as a possible factor affecting the participant's choice of the proper name response. Except for the first person narrator (I), the interlocutor (Bill) and the target referent (Amy), the text did not mention any other human or animal referents.

The narrative was held in the past tense. That is, the events of the main story line (*Bill pushed the plate, I poured milk, I said,* etc.) were in the simple past. The sentences in the direct speech could have a variety of tenses including simple present, past, and future, present perfect, as well as progressive forms. In the intervening thought condition, the tenses were shifted to the past: simple present became past, past and present perfect became past perfect, and *will* became *would*. In addition, if the intervening sequence in the speech condition contained contractions like *I've* or *I'll,* they were spelled out in the thought condition—*I had, I would*—to make the passage sound more narrative and less conversational.

Fillers

The items were mixed with 24 fillers and 8 additional items from another experiment. In some of the fillers, the target referent was maintained in a continuous anaphoric chain, either in the narrative or in the dialogue of the characters, making the pronoun the preferred choice of referring expression in the target sentence. In other fillers, the antecedent was part of a conjunction or a disjunction with a competing referent, e.g. *John and Max* or *John or Max,* where we expected the name *John* to be preferred over the pronoun *he* in the target sentence. The 8 items of the other experiment also contained competing referents, albeit of a different gender, making the pronominal reference more acceptable, but still often dispreferred because of the presence of a competitor.

2.2 Task

We used a forced choice task, where the participants were asked to click on the referring expression, either a name or a pronoun, that they thought should go into the marked gap in the target sentence. They were instructed to read the entire passage before making their choice. Some of the fillers and the additional items from the other experiment (18 in total) were followed by a question about various aspects of the content of the text, to motivate the participants to pay attention to all parts of the text.

2.3 Procedure

The experiment was implemented using Gorilla Experiment Builder (Anwyl-Irvine et al., 2020) and could be accessed online through a standard browser from a computer or a tablet. After giving their informed consent to the participation in the study and to the processing of their data, the participants read the instructions and completed a brief practice session.

The main session consisted of 56 trials (24 items + 24 fillers + 8 items from another experiment), which were presented in a randomized order. Each trial started with a brief presentation of the fixation cross (1 sec). The next screen showed the text and the two response buttons for the referring expressions, the pronoun and the name, underneath the text. The positions of the buttons, whether the pronoun or the name appeared on the right or on the left, were randomized across trials. No reading time limit was implemented; the screen advanced when the participants clicked on one of the response buttons.

In the trials that contained attention questions, the question was presented on the next screen, with two response options yes and no. The left/right position of the response buttons was again randomized.

2.4 Participants

Monolingual English speaking participants from Ireland, the UK and the US without language related disorders were recruited through the Prolific platform (www.prolific.com) and received £7.50 for their participation. In order to raise the odds that our participants had had previous experience reading narrative fiction and were familiar with its conventions, we used Prolific screeners to further narrow the pool to individuals who indicated that literature was one of their hobbies. 80 participants (59 female, 20 male, 1 other) aged 20–73 years (median 47) took part in the experiment. The median duration of the entire session was 31 minutes 41 seconds.

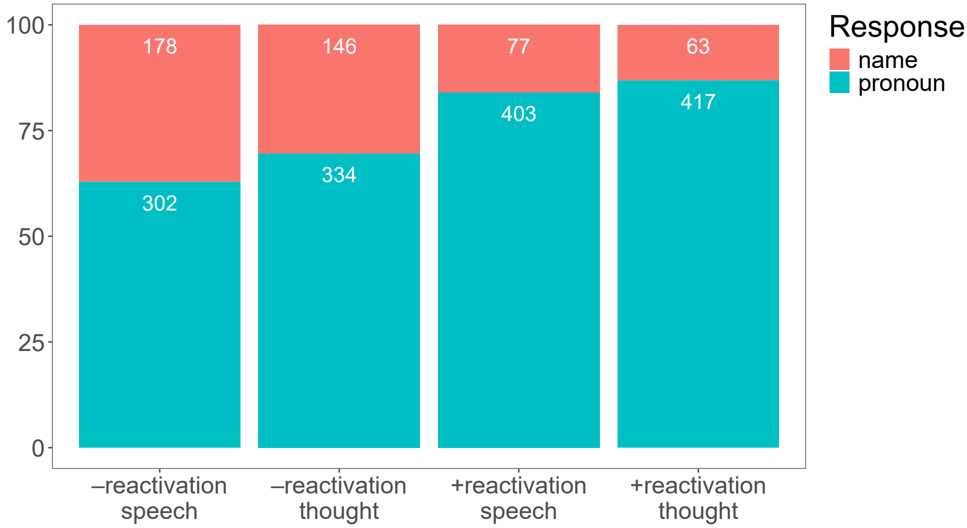

Fig. 1: Percentage of pronoun vs. name responses

3. Results

We collected 1920 observations in the forced choice task. The percentage of pronoun and name responses in each condition is shown in figure 1. Pronoun responses constituted the majority of all responses, between 62.9% and 86.9% depending on the condition, but there were also clear differences. A mixed-effects logistic regression was conducted to examine the effects of reactivation (+ vs. −) and intervening context (speech vs. thought) on the likelihood of a pronoun response. The model included random intercepts for participants and items, but did not include an interaction term. The analysis revealed a significant effect of intervening context, with thought contexts leading to more pronoun responses than speech contexts ($\beta = 0.35$, $SE = 0.12$, $z = 2.88$, $p = .004$). There was also a significant effect of reactivation, such that pronoun responses were more likely following a reactivation of the referent ($\beta = 1.36$, $SE = 0.13$, $z = 10.57$, $p < .001$). The effect of reactivation was substantially more pronounced than that of the intervening speech vs. thought.

To address a reviewer's concern, we also estimated a model including the interaction term for reactivation and intervening speech vs. thought. The interaction was not significant ($\beta = -0.11$, $SE = 0.25$, $z = -0.43$, $p = .67$). Adding the interaction term did not improve model fit, and the main effect estimates were unchanged, suggesting that the more parsimonious specification was adequate.

4. Discussion

The overall preference for pronoun responses that we see in the results is not surprising given that the target referent was the only suitable referent for the pronoun in the entire context and there was no space for ambiguity. It also suggests that participants probably saw the target referent as the global discourse topic of the passage despite the fact that the intervening sequence had a different local topic.

The finding that the pronoun was chosen more frequently in the thought condition than in the speech condition is consistent with our hypothesis that readers take the difference between the narrator-reader CG and the characters' CG into account. Since the referring expression in question occurred in the direct speech in the dialogue between the narrator (as a character in the story) and another character, one would expect that expression to be chosen in accordance with the current state of the characters' CG. Since the other character does not hear the content of the narrator's thoughts in the thought condition, for that character the target referent is still highly activated, as it is the subject of the most recent utterance in their dialogue, while in the speech condition it is separated from its antecedent by several sentences and is therefore less activated. The fact that at least some of the participants were sensitive to that difference suggests that they had been tracking the activation of the target referent in the two CGs separately.

At the same time, the main effect of reactivation suggests that the participants were not very good at keeping the different CGs apart. The sentence before the target sentence did not contain direct speech. In the +reactivation condition, it represented the thought of the narrator that their interlocutor had no access to. Therefore, the mention of the target referent in that sentence reactivated it only in the narrator/reader CG, but not in the characters' CG. However, the target sentence was part of the dialogue of the characters, meaning that the referring expression should be chosen in accordance with the state of the characters' CG. In other words, the target referent was reactivated in the wrong CG, and therefore reactivation should have no influence on the choice of referring expression in the direct speech if readers kept the CGs perfectly apart and the activation of a referent in one CG did not affect its activation in the other. However, what we see is that reactivation made it more likely that participants chose the pronoun response. This suggests that either the activation somehow "spills over" from the narrator/reader CG to the characters' CG, or that readers fail to consider the status of the referent in the characters' CG altogether. The latter would mean that they act egocentrically, paying more attention to the CG they are part of and disregarding the perspective of the characters.

On the one hand, this could be a manifestation of the same phenomenon as defaulting to private information and disregarding the CG in reference tasks (Heller and Brown-Schmidt, 2023). That is, readers have separate representations for multiple CGs, but sometimes fail to consider the CGs that are not their own due to the limitations of the cognitive system. If this is the case, we would expect the preference pattern to change depending on cognitive load or individual psychological characteristics of the participants.

On the other hand, these findings are also consistent with the view that CGs communicate with each other and influence each other within a more abstract representation of context. Like frames, worlds, and points of view, multiple CGs constitute units of representation, whose boundaries dampen activation (cf. Ariel's unity factor), but are not completely impenetrable to it. More research is needed to see how similar the effects of anaphoric relations crossing a CG boundary are to those of crossing a frame or a discourse segment boundary (cf. e.g. Fox, 1987).

However, the results of the presented experiment clearly refute the idea that referential choice is completely determined by the status of the referent in the CG associated with the relevant Kaplanian context. A speech report has a different addressee than the original reported speech, and a referring expression that was optimal for the original addressee, might no longer be optimal for the new one. Referential choice in direct speech is sensitive to that difference rather than being fully dependent on the original speech context.

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